Q3)

a) (i) The loglikelihood is :

$$\mathcal{L}(\beta) = \sum_{i,j \in \{1,2\}} y_{ij} \log(\mu_{ij}) - \mu_{ij}$$

$$= \sum_{ij \in \{1,2\}} y_{ij} (\beta_0 + \beta_1 \mathbb{I}_{\{X_{1,ij} = Y\}} + \beta_2 \mathbb{I}_{\{X_{2,ij} = M\}})$$

$$- \exp(\beta_0 + \beta_1 \mathbb{I}_{\{X_{1,ij} = Y\}} + \beta_2 \mathbb{I}_{\{X_{2,ij} = M\}})$$

- $\dfrac{\partial \mathcal{L}(\beta)}{\partial \beta_0} = \sum_{i,j \in \{1,2\}} y_{ij} - \mu_{ij}$

- $\dfrac{\partial \mathcal{L}(\beta)}{\partial \beta_1} = \sum_{i,j \in \{1,2\}} \mathbb{I}_{\{X_{1,ij} = Y\}} (y_{ij} - \mu_{ij})$

- $\dfrac{\partial \mathcal{L}(\beta)}{\partial \beta_2} = \sum_{i,j \in \{1,2\}} \mathbb{I}_{\{X_{2,ij} = M\}} (y_{ij} - \mu_{ij})$

The equations are :

$$\sum_{i,j \in \{1,2\}} (y_{ij} - \mu_{ij}) = 0 \qquad (*)$$

$$\sum_{i,j \in \{1,2\}} \mathbb{I}_{\{X_{1,ij} = Y\}} (y_{ij} - \mu_{ij}) = 0 \qquad (**)$$

$$\sum_{i,j \in \{1,2\}} \mathbb{I}_{\{X_{2,ij} = M\}} (y_{ij} - \mu_{ij}) = 0 \qquad (***)$$

(ii) (*) $\Rightarrow \sum_{i,j} (y_{ij} - \hat{\mu}_{ij}) = 0$

$\Leftrightarrow \boxed{\sum_{i,j} y_{ij} = \sum_{i,j} \hat{\mu}_{ij}}$

this yields that the total of the observed table is equal to the total of the fitted values.

(**) $\Rightarrow$ for $i = 2$ we have then $I\{X_{1,2,j} = y\} = 1, j = 1,2$

$(**) = \sum_{j=1}^{2} (y_{2j} - \hat{\mu}_{2j}) = 0$

$\Leftrightarrow \boxed{y_{21} + y_{22} = \hat{\mu}_{21} + \hat{\mu}_{22}}$

the result is satisfied by the second row

For the first row, we have $\hat{\mu}_{1\cdot} = y_{\cdot\cdot} - \hat{\mu}_{2\cdot} = y_{\cdot\cdot} - y_{2\cdot} = y_{1\cdot}$

hence $\boxed{y_{11} + y_{12} = \hat{\mu}_{11} + \hat{\mu}_{12}}$

(***) $\Rightarrow$ by using the same reasoning on $(X_2 = M)$:

$\boxed{y_{12} + y_{22} = \hat{\mu}_{12} + \hat{\mu}_{22}}$

Same goes for the first column : $\boxed{y_{11} + y_{21} = \hat{\mu}_{11} + \hat{\mu}_{21}}$

b) (i) the loglikelihood of π_1 and π_2 is:

$$l(\pi_1, \pi_2) = \sum_{i=1}^{2} \sum_{j=1}^{2} y_{ij} \log \pi_{ij} \quad \text{where} \quad \pi_{ij} \overset{\perp}{=} \pi_i \pi_j$$

$$l(\pi_1, \pi_2) = y_{11} \log(\pi_1 \pi_2) + y_{12} \log(\pi_1 (1 - \pi_2))$$

$$+ y_{21} \log((1 - \pi_1)\pi_2) + y_{22} \log((1 - \pi_1)(1 - \pi_2))$$

$$\boxed{l(\pi_1, \pi_2) = y_{1\cdot} \log(\pi_1) + y_{2\cdot} \log(1 - \pi_1) + y_{\cdot 1} \log(\pi_2) + y_{\cdot 2} \log(1 - \pi_2)}$$

. The maximum likelihood estimation of π_1 and π_2:

$$\frac{\partial l(\pi)}{\partial \pi_1} = \frac{y_{1\cdot}}{\pi_1} - \frac{y_{2\cdot}}{1 - \pi_1} = 0 \iff y_{1\cdot} = \pi_1 (y_{1\cdot} + y_{2\cdot})$$

$$\iff \hat{\pi}_1 = \frac{y_{1\cdot}}{n} \quad \text{because} \quad y_{1\cdot} + y_{2\cdot} = n$$

$$\iff \boxed{\hat{\pi}_1 = \frac{1}{n}(y_{11} + y_{12})}$$

by analogy:

$$\boxed{\hat{\pi}_2 = \frac{1}{n}(y_{21} + y_{22})}$$

(ii) For the multinomial model under independance assumption, we have $\hat{\mu}_{ij} = n \hat{\pi}_i \hat{\pi}_j$

X_1 / X_2	M	F
N	$\hat{\mu}_{11} = \frac{y_{1\cdot} y_{\cdot 1}}{n}$	$\hat{\mu}_{12} = \frac{y_{1\cdot} y_{\cdot 2}}{n}$
Y	$\hat{\mu}_{21} = \frac{y_{2\cdot} y_{\cdot 1}}{n}$	$\hat{\mu}_{22} = \frac{y_{2\cdot} y_{\cdot 2}}{n}$

. for the first raw : $\hat{\mu}_{1\cdot} = \hat{\mu}_{11} + \hat{\mu}_{12} = \frac{y_{1\cdot}}{n} \underbrace{(y_{\cdot 1} + y_{\cdot 2})}_{n}$

$= y_{1\cdot} \frac{n}{n}$

finally : $\boxed{\hat{\mu}_{1\cdot} = y_{1\cdot}}$

. second raw : $\hat{\mu}_{2\cdot} = y_{2\cdot} \frac{(y_{\cdot 1} + y_{\cdot 2})}{n} = y_{2\cdot}$

$\boxed{\hat{\mu}_{2\cdot} = y_{2\cdot}}$

. first column : $\hat{\mu}_{\cdot 1} = y_{\cdot 1} \frac{(y_{1\cdot} + y_{2\cdot})}{n} = y_{\cdot 1}$

$\boxed{\hat{\mu}_{\cdot 1} = y_{\cdot 1}}$

. second column : $\hat{\mu}_{\cdot 2} = y_{\cdot 2} \frac{(y_{1\cdot} + y_{2\cdot})}{n} = y_{\cdot 2}$

$\boxed{\hat{\mu}_{\cdot 2} = y_{\cdot 2}}$

c)